

Major Code and Data Migrations for GRNsight, a Web Application for Visualizing Gene Regulatory and Protein-Protein Interaction Network Models

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GRNsight is an open-source web application for visualizing models of gene regulatory networks (GRNs) and protein-protein physical interaction networks (PPIs), either uploaded by the user or retrieved from our back-end database. A gene regulatory network consists of genes, transcription factors, and the regulatory connections between them that govern expression levels of mRNA and proteins. A protein-protein physical interaction network represents the binding relationships between proteins. GRNsight displays graph models where rectangular nodes represent genes or proteins with edges connecting them. While GRNs have directed edges to indicate regulatory relationships with edge thickness indicating regulation strength, PPIs have undirected edges to indicate physical binding. Timecourse gene expression data can be displayed as a heatmap on the individual nodes. As its core, GRNsight integrates a PostgreSQL backend that serves as a centralized repository for molecular interaction and expression data for *Saccharomyces cerevisiae*. Due to the discontinuation of YeastMine, we migrated the data source to AllianceMine, ensuring continuity of high-quality data. In addition, the GRNsight codebase has been migrated from Express JS to a React Vite front-end with Grommet UI styling. Improved state management across the web application and the separation of code into reusable components result in better code maintainability. By combining a durable data backend with a modernized, responsive frontend, GRNsight provides a performant and maintainable framework that allows biologists to gain deeper insights into complex molecular systems. GRNsight Classic v7.4.0 is available at <https://dondi.github.io/GRNsight/>; GRNsight React is available at <https://dondi.github.io/GRNsight/react-thesis-4081>.